POC-1 : **ConfigMap: Volume Mount + Env Var Reference**

Steps:

1.Create ConfigMap app-config with
keys: APP_ENV=production, APP_NAME=inventory-service.

POC > POC-1-ConfigMap-1 > ! configmap.yaml > { } data > abc APP_NAME

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config
data:
  APP_ENV: production
  APP_NAME: inventory-service
```

```
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ ls
configmap.yaml  pod.yaml
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl apply -f configmap.yaml
configmap/app-config created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl get configmap
NAME               DATA   AGE
app-config         2      8s
kube-root-ca.crt   1      30h
special-config     1      27h
```

2 . Create pod config-inspector (toolbox image), that:

- mounts app-config as a volume at /etc/app/config
- also injects APP_ENV as an environment variable via configMapKeyRef

POC-1-ConfigMap-1 > ! pod.yaml > { } spec > [ ] containers > { } 0 > [ ] env > { } 0 > { } valueFrom > { } configMapKeyRef > abc name

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: config-inspector
spec:
  containers:
    - name: toolbox
      image: docker.io/payal_kharat/kubernetes-toolbox:latest
      env:
        - name: APP_ENV
          valueFrom:
            configMapKeyRef:
              name: app-config
              key: APP_ENV
      volumeMounts:
        - name: app-config-volume
          mountPath: /etc/app/config
  volumes:
    - name: app-config-volume
      configMap:
        name: app-config
```

```
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl apply -f pod.yaml
pod/config-inspector created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl get pods
NAME               READY   STATUS    RESTARTS   AGE
config-inspector   1/1     Running   0          21s
pvc-test           1/1     Running   0          32m
```

3 . Verify:

kubectl exec config-inspector -- cat /etc/app/config/APP_ENV

kubectl exec config-inspector -- printenv APP_ENV

```
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl exec config-inspector -- cat /etc/app/config/APP_ENV
productionpayal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl exec config-inspector -- cat /etc/app/config/APP_NAME
inventory-servicepayal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl exec config-inspector -- ls -l /etc/app/config
total 0
lrwxrwxrwx    1 root     root            14 Aug 19 11:30 APP_ENV -> ..data/APP_ENV
lrwxrwxrwx    1 root     root            15 Aug 19 11:30 APP_NAME -> ..data/APP_NAME
```

4.**Expectedresult:**

/etc/app/config/APP_ENV contains production. /etc/app/config/APP_NAME also exists and contains inventory-service (mounted as a whole ConfigMap, both keys appear as files). printenv APP_ENV returns production.
Only APP_ENV exists as an env var — APP_NAME does not, since it was never referenced with configMapKeyRef.

```
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl exec config-inspector -- printenv APP_ENV
production
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl exec config-inspector -- printenv APP_NAME
command terminated with exit code 1
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$ kubectl exec config-inspector -- printenv APP_NAME
command terminated with exit code 1
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-1-ConfigMap-1$
```